

A1. Write a code that uses the vortex-panel method outlined in section 1.6 to compute the lift coefficient, leading-edge pitching-moment coefficient, and quarter-chord pitching-moment coefficient at a specified angle of attack on an airfoil defined by a set of points that are read in from a text file. Use the points as the nodes in the algorithm.

For debugging purposes, the A matrix, B vector, and Gamma vector solutions for a [NACA 2412 airfoil](#) with 10 nodes at zero angle of attack can be viewed [here](#). Results for this example should give the following:

$$\tilde{C}_L = 0.239837889916$$

$$\tilde{C}_{m_{LE}} = -0.111898723302$$

$$\tilde{C}_{m_{c/4}} = -0.0519392508233$$

A2. Use the code developed for A1 to fill out the following table for the [NACA 2412 airfoil](#) with 200 nodes to 5 decimal places. The input to the code should include the following:

```
{
  "geometry" : "2412_200.txt",
  "freestream_velocity" : 10.0,
  "alpha[deg]" : 0.0
}
```

α (deg)	$\tilde{C}_L$	$\tilde{C}_{m_{le}}$	$\tilde{C}_{m_{c/4}}$
-12			
-10			
-8			
-6			
-4			
-2			
0	0.25933	-0.12020	-0.05537
2			
4			
6			
8			
10			
12			
14			
16			

A3. Plot your lift-coefficient solution from A2 in comparison to predictions from thin-airfoil theory and experimental data. The zero-lift angle of attack from thin-airfoil theory is -2.077 degrees. Experimental data can be obtained from [these tables](#). Use the data at a Reynolds number of about 3 million. What do you notice about the results?

A4. Use your airfoil code to fill out the table shown in A2 for a [NACA 2421 airfoil](#) with 200 nodes. Plot your lift-coefficient results in comparison to predictions from thin-airfoil theory and experimental data. The zero-lift angle of

attack from thin-airfoil theory is -2.077 degrees. Experimental data can be obtained from [these tables](#). Use the data at a Reynolds number of about 3 million. What do you notice about the results?

A5. Use your airfoil code to fill out the table shown in A2 for a [NACA 0015 airfoil](#) with 200 nodes. Use XFOIL to fill out the same table for the same airfoil at a Reynolds number of 3 million. Plot the lift-coefficient results from your panel code, XFOIL, and thin-airfoil theory. What do you notice about the results?

Your project submission should include your results for A1-A5 as well as a copy of your airfoil code.